

FastAPI File Upload Error SOP

Objective

Resolve file upload endpoint failures caused by request format, file size, multipart handling, storage errors, or validation logic.

Scope

Applicable to FastAPI-based services, backend APIs, API gateways, DevOps teams, QA teams, and platform teams supporting production, staging, or synthetic incident workflows.

Pre-requisites

1. Access to FastAPI application logs, deployment logs, and API gateway or ingress logs.
2. Access to monitoring dashboards such as Grafana, Prometheus, APM, or cloud monitoring.
3. Permission to view application configuration, environment variables, and deployment status.
4. API testing tool such as curl, Postman, Swagger UI, or automated smoke test scripts.
5. Escalation contact for backend, DevOps, database, and security teams.

Incident Metadata

Field	Value
Ticket ID	T030
Issue	FastAPI file upload error
Category	API
Service	FastAPI
Severity	Low
Status	Resolved
Description	File upload endpoint failing
Expected Resolution	Check file handling logic and request format

Related Log Evidence

No direct matching log entry was present in the provided logs CSV. Use the ticket description, application logs, API gateway logs, and monitoring dashboards to validate this incident in runtime.

Procedure

1. Confirm endpoint, filename, content type, size, and client request format.
2. Check whether request uses multipart/form-data with the expected field name.
3. Review logs for upload parsing, storage, permission, or validation errors.
4. Validate FastAPI UploadFile usage and dependency on python-multipart.
5. Check upload size limits at API gateway, ingress, and application layer.
6. Validate target storage path, permissions, and available disk capacity.
7. Retest upload with a small valid file and then with boundary-size cases.
8. Update ticket with root cause, fixed limit, and validation evidence.

Common Commands

```
curl -i -X GET http://<host>/health
curl -i -X POST http://<host>/<endpoint> -H "Content-Type: application/json" -d @payload.json
```

```
kubectl get pods -n <namespace> | grep fastapi
kubectl logs <fastapi-pod> -n <namespace> --tail=100
kubectl describe pod <fastapi-pod> -n <namespace>
```

Troubleshooting Matrix

Issue	Possible Cause	Resolution
Upload fails	Wrong multipart field	Correct request field and schema
Large files fail	Size limit exceeded	Update approved limits
Parsing error	Missing python-multipart	Install dependency and rebuild
Storage error	Permission or disk issue	Fix storage path and capacity

Best Practices

1. Validate file type and size before processing.
2. Scan or quarantine uploads where required.
3. Use streaming for large uploads.
4. Do not store uploads in ephemeral paths unless intended.

References

[FastAPI Docs](#) | [Internal Logs CSV](#) | [Tickets CSV](#) | [Monitoring Dashboards](#) | [API Gateway Logs](#) | [Backend Runbook](#)